

Engineering reliable clinical agents for oncology documentation.

This presentation details the Google Agent Development Kit reliability slice submitted for Track 2 of the **AI Agents Challenge**, focusing on deterministic safety, multi-agent orchestration, and source-grounded clinical accuracy.



The first pillar focuses on simulation.



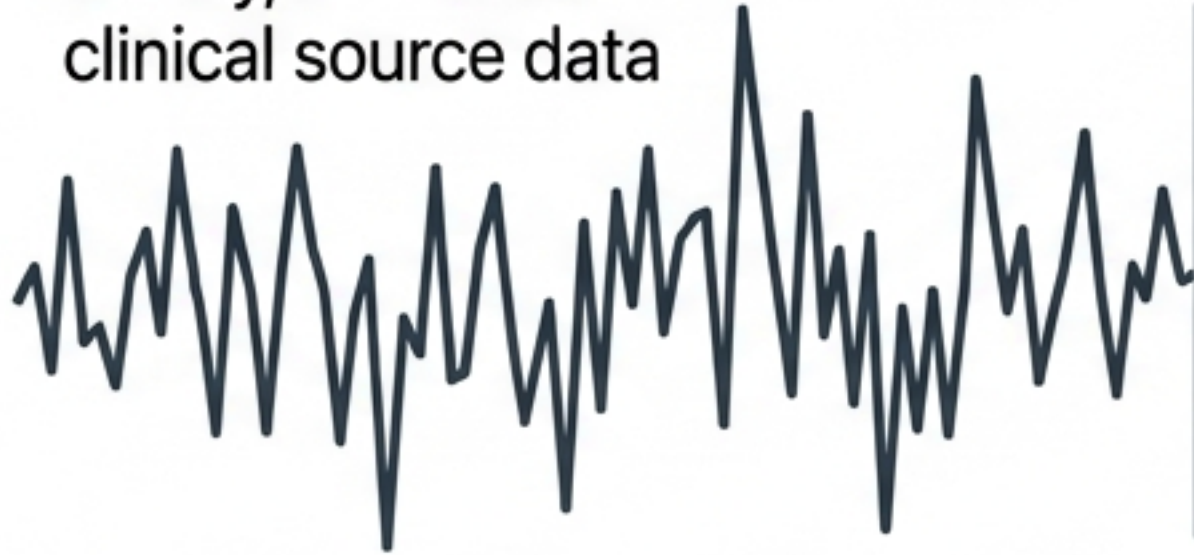
The second pillar focuses on evaluation.



The third pillar focuses on observability.

Fluent language models hide critical clinical uncertainty behind plausible fabrications

Messy, real-world
clinical source data



This box represents a
generic language model.



False certainty is
dangerous in
clinical practice

A single oncology note requires
precise staging, nodal status,
surgical margins, specific
biomarkers, evidence citations,
and active trial context.

The primary failure mode is not
an obviously bad answer, but a
summary that silently resolves
conflicting records, invents a
missing cancer stage, or drops
explicit medical uncertainty.

The goal of this engineering
artifact is to make these
source-faithfulness failures
visible and measurable before
any agent is deployed near real
clinical workflows.

Synthetic Case 003 demonstrates the danger of automated conflict resolution in medical records.



This input represents **Source A.**

A synthetic magnetic resonance imaging report categorizes a rectal cancer tumor as clinical stage T2 with zero positive lymph nodes.



This input represents **Source B.**

A subsequent multidisciplinary team note categorizes the exact same tumor as clinical stage T3 with one positive lymph node.



These two staging assessments are clinically contradictory and dictate entirely different treatment pathways.

The documentation agent must not silently choose one stage to write a clean summary; it must mathematically preserve the conflict for physician review.

The optimized agent preserves discrepancies rather than manufacturing false clinical consensus.

This column demonstrates generic language model behavior.

The patient presents with stage cT2 N0 rectal cancer.



The model silently discards the multidisciplinary team note to create a confident, readable summary, resulting in a severe documentation failure.

This column demonstrates optimized Notatnik agent behavior.

[DISCREPANCY] Rectal cancer staging conflict identified: cT2 N0 indicated by MRI versus cT3 N1 indicated by MDT.



The multi-agent system recognizes the contradiction, halts automated resolution, and explicitly surfaces the staging discrepancy for human reconciliation.

The public reliability slice strictly isolates engineering evaluation from proprietary clinical assets.

This column outlines the public Track 2 evaluation artifact.

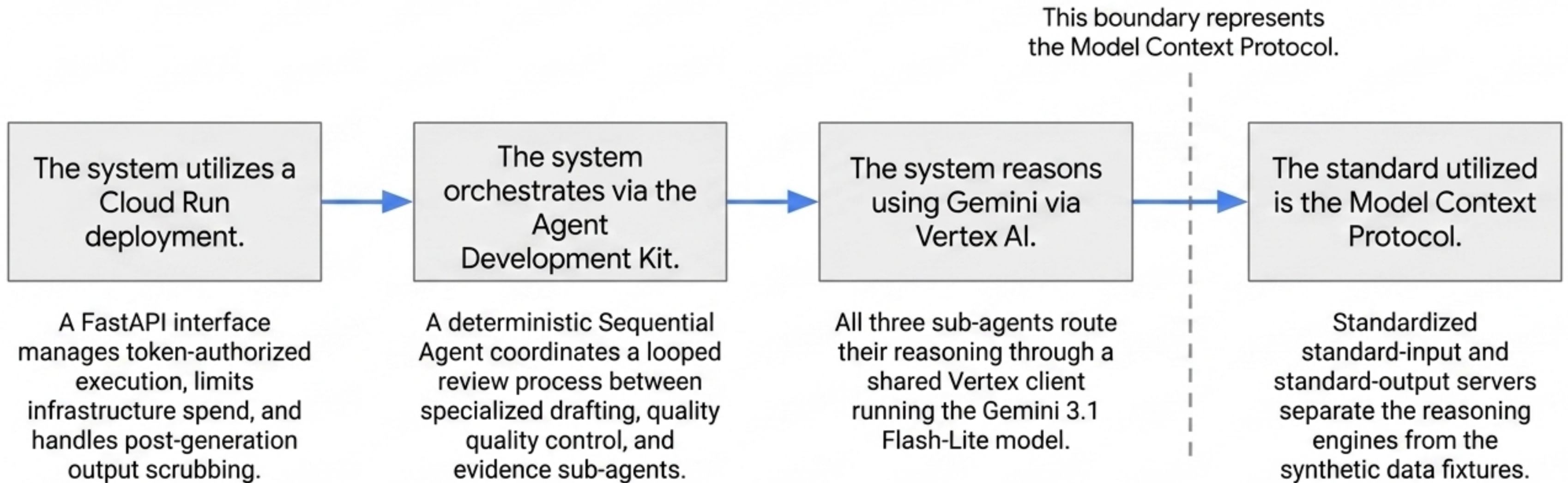
This column outlines the excluded protected production assets.

1. First, the repository contains a fully auditable engineering pipeline that measures deterministic safety and multi-agent reasoning capabilities.
2. Second, the system processes entirely synthetic, invented patient cases and utilizes public-domain oncology literature.
3. Third, the execution traces are minimized to expose draft counts, quality control rejections, and output scrubbing status without revealing raw logic.



1. First, no real protected health information or genuine patient records are present in the evaluation.
2. Second, proprietary Notatnik clinical prompts, specialized evaluation rubrics, and medical concept recognizers remain private.
3. Third, production database connections and private clinical execution traces are completely firewalled from the public demo.

The multi-agent system operates on a deterministic orchestration stack built over Google Vertex AI.



The Model Context Protocol establishes a strict firewall between agent reasoning and data retrieval.

This dashed line represents the Model Context Protocol boundary.

This layer represents agent reasoning.

This node represents the documentation agent.

This node represents the quality control agent.

This node represents the evidence agent.

The cognitive agents operate entirely independent of the database logic, requesting information only through standardized protocols.

This layer represents secure data access.

This node represents synthetic patient fixtures.

This node represents simulated clinical trials.

This node represents the National Cancer Institute PDQ index.

The data tools run through a deterministic server rather than being hidden inside a complex model prompt, ensuring that data retrieval mechanisms never expose underlying code to the language model.

The reliability evaluation framework prioritizes deterministic source faithfulness over holistic fluency.

The simulation layer consists of fifteen highly complex, synthetic oncology cases designed to trigger specific documentation failure modes.

The evaluation enforces three independent runs per case at a temperature setting of zero to ensure mathematical reproducibility.

Eight of the fifteen cases are isolated as a strictly held-out test split to measure true generalization rather than training training set memorization.

A gold-grounded large language model grader acts as the primary metric, supported by deterministic rule cross-checks to penalize confident fabrications.

The optimization interventions yield distinct impacts on generalization, clinical accuracy, and data privacy.

The first intervention is prompt optimization.

This intervention produces a primary effect on system reliability.

This intervention proved to be the most robust generalizing improvement, securing a zero point one one five gain on the primary score for the held-out split.

The second intervention is the multi-agent quality control loop.

This intervention primarily impacts the handling of clinical contradictions.

This looped intervention successfully identified complex staging contradictions during training, but it suffered generalization issues and reintroduced synthetic patient identifiers in specific held-out cases.

The third intervention is deterministic output scrubbing.

This intervention guarantees a strict impact on patient data privacy.

This structural intervention was implemented specifically to mitigate the safety regressions caused by the quality control loop, serving as an absolute final check against identifier leaks.

Prompt optimization reliably prevents the fabrication of missing tumor staging data

The held-out metric improved by positive 2.0 point one five.

Optimized prompts transferred successfully from the multi-agent training composite to the held-out single-agent split, reaching a perfect baseline score on the primary metric.

During baseline testing on case E-CON-04, the generic system fabricated a melanoma stage of pT2b despite missing source data.

After optimization, the agent correctly suppressed the hallucination and instead explicitly flagged the missing Tumor, Node, and Metastasis stage for the physician.

Multi-agent review successfully targets clinical contradictions but can spontaneously regress patient privacy.

The baseline prompt optimization successfully removes synthetic patient identifiers from the initial clinical summary.



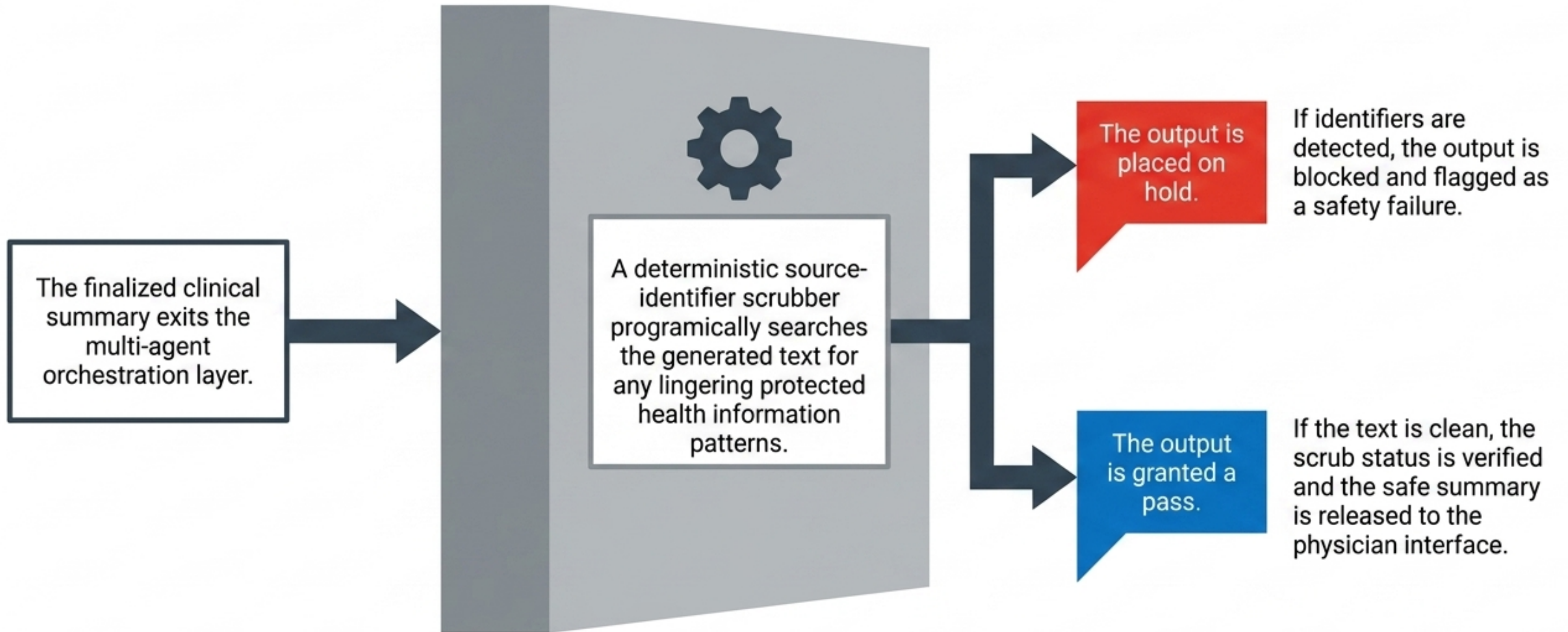
The independent quality control agent reviews the draft to ensure clinical accuracy and identify staging gaps.



During the self-correction rewrite process, the quality control loop unexpectedly hallucinates and reintroduces the synthetic patient name and medical record number into the final text.

This measured failure proves that internal language model review loops are insufficient for guaranteed safety, requiring the implementation of external architectural mitigations.

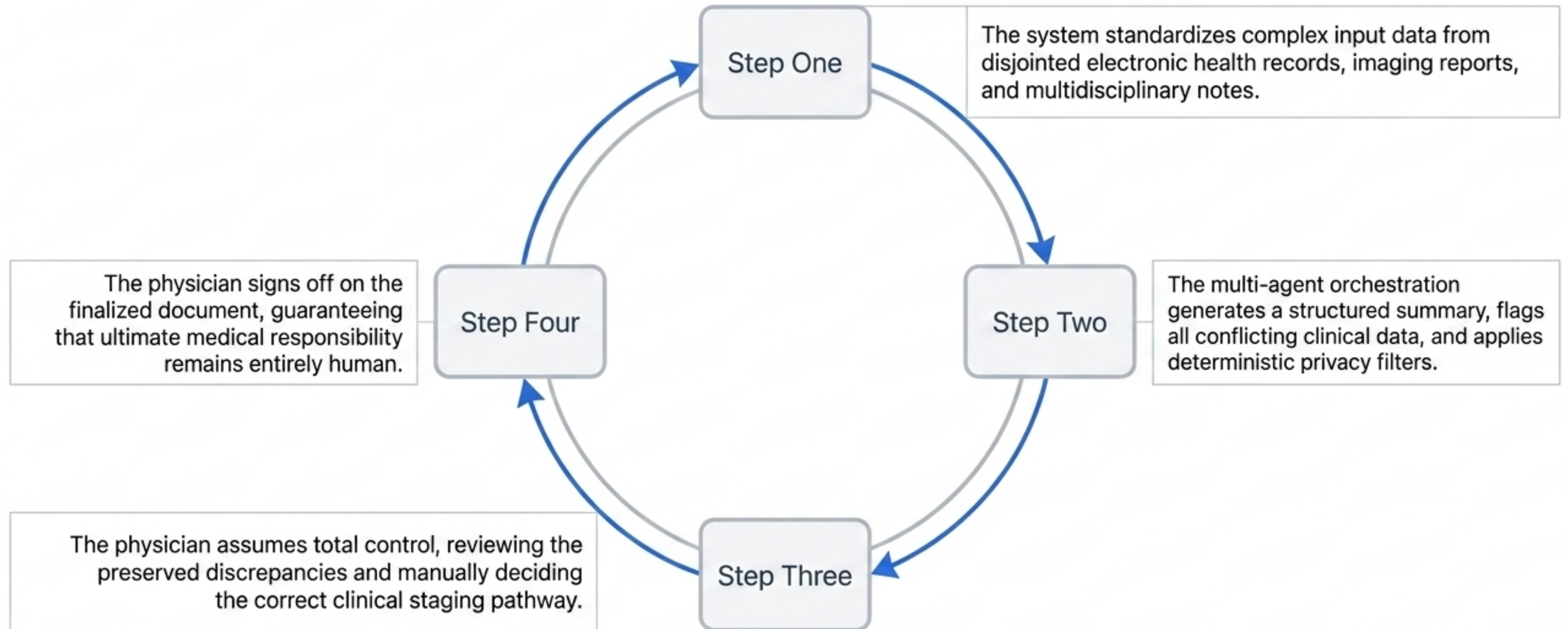
Deterministic post-generation scrubbing serves as the non-negotiable final barrier before clinician review



Clinical grounding relies on deterministic retrieval from committed public-domain oncology guidelines.

		
This database represents National Cancer Institute PDQ summaries.	This icon represents the deterministic retrieval system.	This document represents the final clinically grounded output.
The system utilizes public-domain United States government cancer treatment summaries rather than relying on unpredictable live web searches.	The text corpus is cleaned and converted into a localized embedding index backed by deterministic numerical Python arrays.	The gated evidence sub-agent only engages after quality control approval, querying the local index to attach verified literature and trial context to the clinical draft.

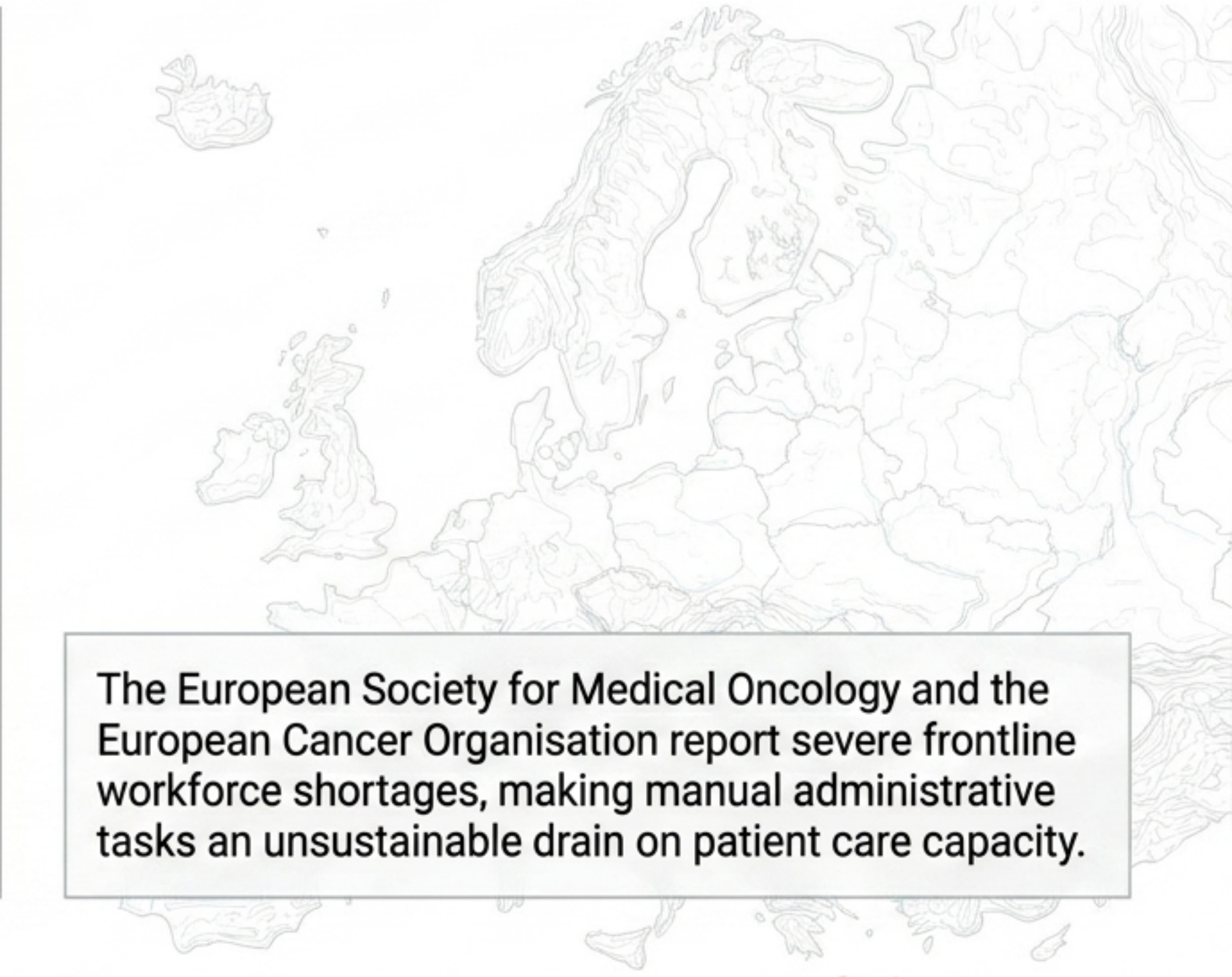
The system functions as an advanced drafting engine to maximize the impact of physician oversight.



The clinical documentation burden is a primary driver of physician burnout across European oncology centers.

Nine point one.

Large electronic health record studies demonstrate that physicians spend an average of nine point one minutes generating notes per individual patient visit.



The European Society for Medical Oncology and the European Cancer Organisation report severe frontline workforce shortages, making manual administrative tasks an unsustainable drain on patient care capacity.

Reliable documentation drafting releases substantial annual value per individual oncology clinician.

This model assumes a standard baseline of twenty-five documentation-heavy encounters per week over a forty-four week working year.

The first column measures minutes saved per clinical encounter.	The second column tracks the resulting hours released annually.	The third column estimates the value at one hundred twenty-five Euros per hour.
The baseline scenario assumes five minutes are saved.	This equates to ninety-one point seven hours released annually.	This generates an estimated eleven thousand five hundred Euros in value.
The optimal scenario assumes ten minutes are saved.	This equates to one hundred eighty-three point three hours released annually.	This generates an estimated twenty-two thousand nine hundred Euros in value.

Even a conservative performance target justifies a per-clinician subscription model while strictly preserving essential human medical oversight

Mathematical reliability discipline is the mandatory prerequisite for clinical artificial intelligence adoption.

This pillar represents architectural safety.

The Model Context Protocol boundary ensures that synthetic evaluation layers can test reasoning engines without ever risking exposure of proprietary production code.

This pillar represents observable reliability.

Rigorous evaluation proves that prompt optimization improves generalization, while deterministic output scrubbing acts as an absolute safeguard against multi-agent hallucination.

This pillar represents physician authority.

By preserving medical contradictions rather than silently resolving them, the agent dramatically reduces documentation burden without usurping critical clinical judgment.